

Lecture 5:

Qubits collectively:

Composition and entanglement

Luís Soares Barbosa
www.di.uminho.pt/~lsb/



Universidade do Minho



INESC TEC



UNU

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Building larger states from smaller

Recall that operator U in the no-cloning theorem acts on a 2-dimensional state, i.e. over the composition of two qubits.

What does composition mean?

The composition postulate

The state space of a combined quantum system is the tensor product $V \otimes W$ of the state spaces V and W of its components.

Composing quantum states

State spaces in a **quantum** system combine through **tensor**: $\otimes$

n m -dimensional vectors $\rightsquigarrow$ a vector in m^n -dimensional space

i.e. the state space of a quantum system grows exponentially with the number of particles

Example

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \otimes \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix} = \begin{bmatrix} ad \\ ae \\ af \\ bd \\ be \\ bf \\ cd \\ ce \\ cf \end{bmatrix}$$

Composing quantum states

Tensor $V \otimes W$

- $B_{V \otimes W}$ is a set of elements of the form $|v_i\rangle \otimes |w_j\rangle$, for each $|v_i\rangle \in B_V$, $|w_j\rangle \in B_W$ and $\dim(V \otimes W) = \dim(V) \times \dim(W)$
- $(|u_1\rangle + |u_2\rangle) \otimes |z\rangle = |u_1\rangle \otimes |z\rangle + |u_2\rangle \otimes |z\rangle$
- $|z\rangle \otimes (|u_1\rangle + |u_2\rangle) = |z\rangle \otimes |u_1\rangle + |z\rangle \otimes |u_2\rangle$
- $(\alpha|u\rangle) \otimes |z\rangle = |u\rangle \otimes (\alpha|z\rangle) = \alpha(|u\rangle \otimes |z\rangle)$
- $\langle (|u_2\rangle \otimes |z_2\rangle) | (|u_1\rangle \otimes |z_1\rangle) \rangle = \langle u_2 | u_1 \rangle \langle z_2 | z_1 \rangle$

Composing quantum states

Clearly, every element of $V \otimes W$ can be written as

$$\alpha_1(|v_1\rangle \otimes |w_1\rangle) + \alpha_2(|v_2\rangle \otimes |w_1\rangle) + \cdots + \alpha_{nm}(|v_n\rangle \otimes |w_m\rangle)$$

Example

A basis for $V \otimes W$ can be written in terms of the computational basis:

$$\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$$

Thus, the tensor of $\alpha_1|0\rangle + \alpha_2|1\rangle$ and $\beta_1|0\rangle + \beta_2|1\rangle$ is

$$\alpha_1\beta_1|0\rangle \otimes |0\rangle + \alpha_1\beta_2|0\rangle \otimes |1\rangle + \alpha_2\beta_1|1\rangle \otimes |0\rangle + \alpha_2\beta_2|1\rangle \otimes |1\rangle$$

i.e., in a simplified notation,

$$\alpha_1\beta_1|00\rangle + \alpha_1\beta_2|01\rangle + \alpha_2\beta_1|10\rangle + \alpha_2\beta_2|11\rangle$$

Bases

The computational basis for a vector space

$$\underbrace{V \otimes V \otimes \dots \otimes V}_n$$

corresponding to the composition of n qubits (each living in V) is the set

$$\begin{aligned} & \{ \underbrace{|0\rangle \dots |0\rangle |0\rangle}_n, \underbrace{|0\rangle \dots |0\rangle |1\rangle}_n, \underbrace{|0\rangle \dots |1\rangle |0\rangle}_n, \dots, \underbrace{|1\rangle \dots |1\rangle |1\rangle}_n \} \\ \stackrel{\text{abv}}{=} & \{ \underbrace{|0 \dots 00\rangle}_n, \underbrace{|0 \dots 01\rangle}_n, \underbrace{|0 \dots 10\rangle}_n, \dots, \underbrace{|1 \dots 11\rangle}_n \} \end{aligned}$$

which may be written in a compressed (decimal) way as

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle, \dots, |2^n - 1\rangle\}$$

Bases

The **computational basis** for a two qubit system would be

$$\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$$

with

$$|0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad |2\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad |3\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Bases

There are of course other bases ... besides the **standard** one, e.g.

The Bell basis

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Compare with the Hadamard basis for the single qubit systems

Representing multi-qubit states

Any unit vector in a 2^n -Hilbert space represents a possible n -qubit state, but for

... a certain level of redundancy

- As before, vectors that differ only in a **global phase** represent the **same** quantum state
- but also the **same phase factor in different qubits** of a tensor product represent the **same** state:

$$|u\rangle \otimes (e^{i\phi}|z\rangle) = e^{i\phi}(|u\rangle \otimes |z\rangle) = (e^{i\phi}|u\rangle) \otimes |z\rangle$$

Actually, phase factors in qubits of a single term of a superposition can always be factored out into a coefficient for that term, i.e.
phase factors distribute over tensors

Representing multi-qubit states

- Relative phases still matter (of course!)

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \text{ differs from } \frac{1}{\sqrt{2}}(e^{i\phi}|00\rangle + |11\rangle)$$

even if

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(e^{i\phi}|00\rangle + e^{i\phi}|11\rangle) = \frac{e^{i\phi}}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- The complex **projective space** of dimension 1 (depicted in the **Block sphere**) generalises to higher dimensions, although in practice linearity makes Hilbert spaces easier to use.

Separable and entangled states

State $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$ can be decomposed as $|0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

But this not true for $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ which is called an **entangled** state.

Actually, most states are entangled:

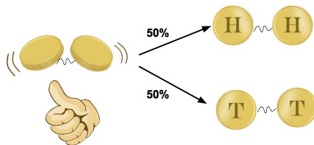
- To express a composition of n qubits requires $2n$ complex numbers (two amplitudes per qubit)
- But how many states $|b_1, b_2, \dots, b_n\rangle$ are there in a Hilbert space dimension n ?

Each of them is labelled by a binary string of length n , and there are 2^n such strings

$$2^n \gg 2n$$

Entanglement

Entanglement express **correlations** between **subsystems**:



*"I would not call it one but rather **the characteristic** trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought"*

(Erwin Schrödinger, 1935)

Indeed, for **better** and **worse** ... it comes from **interaction** between parts of a system ...

Example: An entangled state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

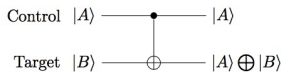
Actually, to make $|\Phi^+\rangle$ equal to

$$(\alpha_1|0\rangle + \beta_1|1\rangle) \otimes (\alpha_2|0\rangle + \beta_2|1\rangle) = \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \beta_1\alpha_2|10\rangle + \beta_1\beta_2|11\rangle$$

would require that $\alpha_1\beta_2 = \beta_1\alpha_2 = 0$ which implies that either

$$\alpha_1\alpha_2 = 0 \text{ or } \beta_1\beta_2 = 0$$

The *CNOT* gate entails such an interaction



In general: Conditional X

$$CNOT|x\rangle|y\rangle = |x\rangle|x \oplus y\rangle$$

Particular case: The 'spy' gate

$$CNOT|x\rangle|0\rangle = |x\rangle|x\rangle$$

CNOT: conditional X

This paradigmatic two-qubit gate acts on the standard basis for a 2-qubit system flipping the second bit if the first bit is 1 and leaving it unchanged otherwise.

$$\begin{aligned} \text{CNOT} &= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X \\ &= |0\rangle\langle 0| \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) + |1\rangle\langle 1| \otimes (|1\rangle\langle 0| + |0\rangle\langle 1|) \\ &= |00\rangle\langle 00| + |01\rangle\langle 01| + |11\rangle\langle 10| + |10\rangle\langle 11| \end{aligned}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

A source of entanglement

CNOT generates **entanglement** e.g.

$$\begin{aligned} \text{CNOT}(H \otimes \text{Id})|00\rangle &= \text{CNOT}\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle\right) \\ &= \text{CNOT}\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \end{aligned}$$

Being its own inverse, also takes an entangled to an unentangled one.

Note that **entanglement** is not a local property in the sense that transformations that act separately on two or more subsystems cannot affect the entanglement between those subsystems:

$$(U \otimes V)|v\rangle \text{ is entangled iff } |v\rangle \text{ is}$$

A source of entanglement

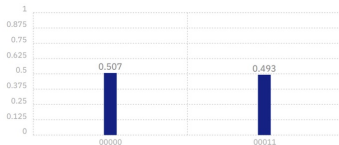
The Bell states

$$CNOT(H \otimes Id)|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$CNOT(H \otimes Id)|01\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$CNOT(H \otimes Id)|10\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$CNOT(H \otimes Id)|11\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$



They are **maximally entangled** states and form a **basis** for a two-dimension Hilbert space

CNOT is unitary and is its own inverse, and **cannot be decomposed into a tensor product of two 1-qubit transformations**

Shared randomness

Let's now discuss how **entanglement** can be used as (another) **computational resource**.

A protocol

Let two agents, Alice and Bob, in different (possibly distant) locations, share a sequence of **Bell pairs**

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

When Alice measures her qubit in the pair, she immediately becomes aware of what the result of a measurement by Bob will be when he decides to make it: the global state has already collapsed.

Shared randomness

- Of course there is **no communication** at all involved in this protocol as the value measured by Alice is perfectly random and to communicate one needs first to know what is to be communicated.
- However, by repeating this process for all shared Bell pairs **two identical sequences of completely random binary values** will be produced in both locations, which is an important resource in e.g. cryptographic protocols, For example,

0 1 1 0 ... 0

0 1 1 0 ... 0

Teleportation

Aim: to transmit, using **two classical bits**, the state of **a single qubit**.

Surprisingly, the protocol

- shows that two classical bits suffice to communicate a qubit, which has an **infinite number of configurations**
- provides a mechanism for the transmission of an unknown quantum state, **in spite of the no-cloning theorem**

Note that the **original state cannot be preserved** (precisely because of the no-cloning result), which motivates the name of the protocol ...

Teleportation

Strategy:

- Alice and Bob share a **Bell state**

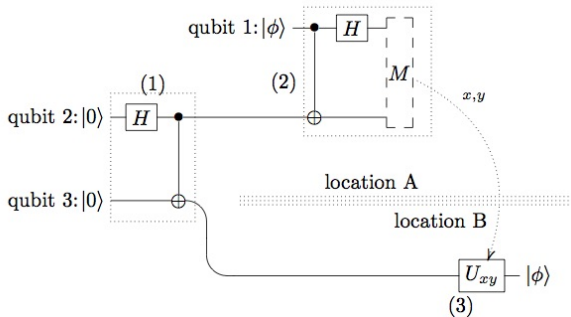
$$|r\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$$

created ahead in time when both qubits were together and could be made to interact to produce an entangled state (through e.g. a *CNOT* gate).

- This entangled state becomes a **resource** which remains available when one qubit is taken away from the other.
- Then Alice encodes the unknown qubit with her part of the Bell pair, measures and transmits the (classical) result of the measurement.
- Finally, Bob **decodes** its part of the Bell pair based on the (classical) information received

Teleportation

Implementation:



The Bell pair

is the real **resource**

also named EPR pair after Einstein, Podolsky, and Rosen, from the *hidden-variable* controversy

Teleportation

Alice

... has a qubit whose state $|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$ she does not know, but wants to send to Bob through classical channels.

The starting point is the 3-qubit state **after** stage (1) whose first 2 qubits are controlled by Alice and the last by Bob:

$$\begin{aligned} |\phi\rangle \otimes |r\rangle &= \frac{1}{\sqrt{2}} (\alpha|0\rangle \otimes \overbrace{(|00\rangle + |11\rangle)}^{\text{entangled}} + \beta|1\rangle \otimes \overbrace{(|00\rangle + |11\rangle)}^{\text{entangled}}) \\ &= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle) \end{aligned}$$

Teleportation

Alice

... then she applies $CNOT \otimes I$ and $H \otimes I \otimes I$ to obtain

$$\begin{aligned} & (H \otimes I \otimes I)(CNOT \otimes I)(|\phi\rangle \otimes |r\rangle) \\ &= (H \otimes I \otimes I) \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle) \\ &= \frac{1}{2} (\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)) \\ &= \frac{1}{2} (|00\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |01\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + \\ &\quad + |10\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) + |11\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)) \end{aligned}$$

Teleportation

Alice

Alice **measures** the first two qubits and obtains one of the four standard basis states, $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$, with equal probability.

Depending on the result of her measurement, the state of Bob's qubit is **projected** to

$$\alpha|0\rangle + \beta|1\rangle, \alpha|1\rangle + \beta|0\rangle, \alpha|0\rangle - \beta|1\rangle, \alpha|1\rangle - \beta|0\rangle$$

Then, Alice **sends** the result of her measurement as two classical bits to Bob.

After these transformations, **crucial information** about the original state $|\phi\rangle$ is contained in Bob's qubit, Alice's being **destroyed** ...

Teleportation

Bob

When Bob receives the two bits from Alice, he knows how the state of his half of the entangled pair compares to the original state of Alice's qubit.

Bob can **reconstruct** the original state of Alice's qubit, $|\phi\rangle$, by applying the appropriate decoding transformation to his qubit, originally part of the entangled pair.

Bits received	Bob's state	Transformation to decode
00	$\alpha 0\rangle + \beta 1\rangle$	I
01	$\alpha 1\rangle + \beta 0\rangle$	X
10	$\alpha 0\rangle - \beta 1\rangle$	Z
11	$\alpha 1\rangle - \beta 1\rangle$	Y

After **decoding**, Bob's qubit will be in the state $|\phi\rangle$

Dense coding

Aim: encode and transmit **two classical bits** with **one qubit** and a shared Bell pair.

This result is surprising, since only one bit can be extracted from a qubit

The idea is that, since entangled states can be distributed ahead of time, only one qubit needs to be physically transmitted to communicate two bits of information.

Let Alice (Bob) be sent and operate the first (second) qubit of the **EPR pair**

$$|r\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Dense coding

Alice

wishes to transmit the state of two classical bits **encoding** one of the numbers 0 through 3. Depending on this number, Alice performs one of the Pauli transformations on her qubit of the entangled pair $|r\rangle$, and **sends her qubit** to Bob.

	Transformation	New state
0	$(I \times I) r\rangle$	$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$
1	$(X \times I) r\rangle$	$\frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$
2	$(Z \times I) r\rangle$	$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$
3	$(Y \times I) r\rangle$	$\frac{1}{\sqrt{2}}(- 10\rangle + 01\rangle)$

Dense coding

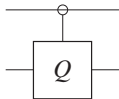
Bob

decodes the information applying a *CNOT* gate to the two qubits of the entangled pair and then *H* to the first qubit:

$$\begin{aligned}
 \text{CNOT} &\longrightarrow \begin{bmatrix} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \\ \frac{1}{\sqrt{2}}(|11\rangle + |01\rangle) \\ \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) \\ \frac{1}{\sqrt{2}}(-|11\rangle + |01\rangle) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\ \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle) \otimes |1\rangle \\ \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |0\rangle \\ \frac{1}{\sqrt{2}}(-|1\rangle + |0\rangle) \otimes |1\rangle \end{bmatrix} \\
 H \otimes I &\longrightarrow \begin{bmatrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{bmatrix}
 \end{aligned}$$

Bob then measures the two qubits in the standard basis to obtain the 2-bit binary encoding of the number Alice wished to send

Controlled Q-gates



$$C_Q|0\rangle|\varphi\rangle = |0\rangle|\varphi\rangle$$

$$C_Q|1\rangle|\varphi\rangle = |1\rangle Q|\varphi\rangle$$

$$C_Q = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Q$$

corresponding to the following matrix in the standard basis:

$$C_Q = \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix}$$

CCNOT or Toffoli gate

A 3-bit gate corresponding to **controlled CNOT**. If the first two bits are in the state $|1\rangle$ applies X the third bit, else it does nothing:

$$|q_1 q_2 q_3\rangle \mapsto |q_1 q_2, q_3 \oplus (q_1 \wedge q_2)\rangle$$

In matrix form,

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Universal set of gates?

Is there a universal set of quantum gates?

In general **no**: there are uncountably many quantum transformations, and a finite set of generators can only generate countably many elements. However, it is possible for **finite sets of gates** to generate **arbitrarily close approximations to all unitary transformations**.

Definitions

- The **error** in approximating U by V is

$$Er(U, V) = \max_{|\phi\rangle} |(U - V)|\phi\rangle|$$

- An operator U can be **approximated to arbitrary accuracy** if for any positive ϵ there exists another unitary transformation V st $Er(U, V) \leq \epsilon$.
- A set of gates is **universal** if for any integer $n \geq 1$, any n -qubit unitary operator can be approximated to arbitrary accuracy by a quantum circuit using only gates from that set.

Universal set of gates?

Some examples

- The set $\{H, T\}$ is universal for 1-gates.
- The set $\{H, T, CNOT\}$ is a universal set of gates.

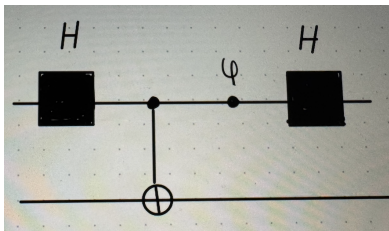
How efficient is an approximation?

To approximate an unitary transformation encoding some specific computation, one would expect to use a number of gates from the universal set which is **polynomial** in the number of qubits and the inverse of the quality factor ϵ .

Main result: theorem of **Solovay-Kitaev**

Entanglement and Interference

Let's combine the **interference** and the **entanglement** patterns:



and apply it to $|00\rangle$:

$$(H \otimes I)(P_\phi \otimes I)CNOT(H \otimes I)|00\rangle$$

Entanglement and Interference

$$(H \otimes I)(P_\varphi \otimes I) \textcolor{red}{CNOT}(H \otimes I) |00\rangle$$

- Before P_φ the system is in the first Bell state; after it becomes

$$\frac{1}{\sqrt{2}}(|00\rangle + e^{i\varphi}|11\rangle)$$

- The second H closes superposition yielding

$$\frac{1}{2}(|00\rangle + |10\rangle + e^{i\varphi}|01\rangle - e^{i\varphi}|11\rangle) = |0\rangle \left(\frac{|0\rangle + e^{i\varphi}|1\rangle}{2} \right) + |1\rangle \left(\frac{|0\rangle - e^{i\varphi}|1\rangle}{2} \right)$$

Entanglement and Interference

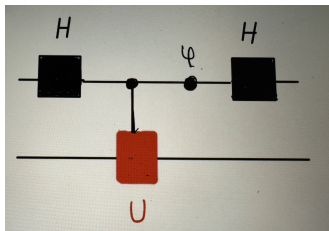
An easy calculation shows that the probabilities of measuring 0 or 1 are $\frac{1}{2}$ in each case.

Interference (via P_φ) was **cancelled** by entanglement: Indeed the second qubit **learnt** the first one, becoming aware of which path in the superposition was taken.

The *CNOT* gate, the **spy** gate, **destroyed** interference

Entanglement and Interference

Let's go general: replacing $CNOT$ by a controlled C_U for an arbitrary unitarian U :



$$(H \otimes I)(P_\phi \otimes I)C_U(H \otimes I)|00\rangle$$

Entanglement and Interference

$$C_U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$$

- If the control qubit is $|0\rangle$, the target qubit remains $|0\rangle$
- if not it is **rotated** from $|0\rangle$ to $U|0\rangle$

As before, if $U = CNOT$ the rotation is π : the new state becomes orthogonal to the previous one.

The second qubit acquired **full knowledge** about the first one.

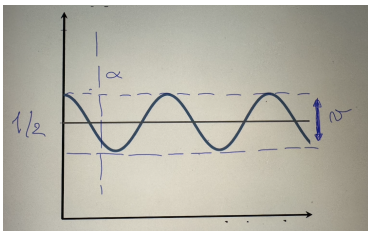
In general, however, the acquired knowledge may be **imperfect**:

$$\langle 0|U|0\rangle = ve^{-i\alpha}$$

recall: the inner product measures the similarity between states

Entanglement and Interference

$$\langle 0|U|0\rangle = ve^{-i\alpha}$$



Special cases

$\langle 0|X|0\rangle = 0$ Interference completely **suppressed**: the curve becomes **flat** at $\frac{1}{2}$.

$\langle 0|I|0\rangle = 1$ **Full** interference: probabilities oscillate as $\cos \frac{\phi}{2}$.

Entanglement as an adversary

Suppose our system corresponds to the first qubit and consists of the typical interference circuit: $HP_\phi H$.

Interacting with the **environment** it becomes **open** and easily goes into an entanglement situation, e.g. as in the previous of the diagram.

However, the system has now no control over the environment.

This may lead to reduction, of even suppression, of interference and therefore to the **collapse of quantum computational power**.

Remember the fridge?



Decoherence of our quantum system starts here ...

... but is a much longer story